The Irrelevance Result says that even if we have the right benchmark for the asset owner, the benchmark is irrelevant for the fund manager's actions when she is paid using a linear contract. But we can give the agent a different benchmark.

3.3. OPTIMAL DESIGN OF BENCHMARKS

One way to break the Irrelevance Result is to use a *smarter* benchmark. The benchmarks in the simple Stoughton-Admati-Pfleiderer setups are simple, *static* asset class benchmarks, like a traditional S&P 500 index. If we insist on using a linear contract, then it is possible to hire the best agent, and get the best agent to optimally put in effort, by changing the static benchmark to a smarter *dynamic* benchmark. This is what Ou-Yang (2003) shows.[10] Ou-Yang creates an *active* benchmark where the number of shares in each asset varies over time, rather than using a passive index where the number of shares invested in each asset is fixed.

The factor benchmarks, which chapter 14 considers, do precisely this. In dynamic benchmarks, the number of shares varies over time and in a factor benchmark the time variation is engineered to optimally maximize exposure to factor risk. Although this is not exactly the same optimal benchmark that Ou-Yang's setup would predict (and Ou-Yang's optimal benchmark can be difficult to compute), the overall concept is that we should move toward giving delegated portfolio managers smarter dynamic benchmarks if we retain linear payment structures. At the very least, we know from the Irrelevance Result that the static benchmark does not create any value for the asset owner.

Factor benchmarks also pick up common or aggregate shocks over which agents have no control and then relative performance evaluations offer better risk-sharing because they filter out the common shocks and reward or penalize the actions over which the agent does have control. Factor benchmarks, if chosen correctly, also make it harder for agents to "fake skills" that masquerade as excess returns (or alpha; see chapter 10) relative to passive indexes.[11] In particular, investment strategies with highly skewed payoffs, like those exposed to volatility risk, are best measured with dynamic factors that explicitly account for nonlinear risk.

The basic setting in the Irrelevance Result is a one-shot delegation model, where the asset owner finds portfolio managers over a single period. In that sense, it is similar to the Capital Asset Pricing Model. The single period is often not a constraint, as chapter 4 shows, because dynamic problems are simply a series of

[10] This paper is also remarkable because it is the first principal-agent model where the agent's action affects both the return (drift) and the risk (diffusion) simultaneously, which is the key issue in the delegated portfolio management problem, and Ou-Yang succeeds in deriving an optimal linear contract breaking the Irrelevance Result.

[11] Makarov and Plantin (2012) argue that is optimal for agents to fake alpha by taking on tail risk that cannot be correctly measured by static indexes.